

Homework Assignment 1

1. Nitrogen gas has molar mass of 28 kg/kmol. At temperature 1100 K, it has specific enthalpy of 885 kJ/kg and constant pressure specific heat of 1188 J/kg-K.
 - (a) Calculate the specific gas constant of nitrogen
 - (b) Find molar specific enthalpy of nitrogen at 1100 K.
 - (c) Find the molar, constant pressure specific heat and the specific heat ratio of nitrogen at 1100 K.
2. The specific gravities of liquid oxygen (an oxidizer) and RP-1 (a fuel) are, respectively, 1.14 and 0.81, respectively. Calculate their densities.
3. The chemical formula of nitrogen tetroxide (an oxidizer) is N_2O_4 . Beginning with the masses of the constituent atoms, find its molar mass. Repeat for UDMH (a fuel), which has the chemical formula $(\text{CH}_3)_2\text{NNH}_2$.
4. Combustion gases have molar mass of 30, and constant pressure specific heat given by the equation $c_p = 0.959 + 1.16(10^{-4})T + 3.65(10^{-8})T^2$ (T in K, c_p in kJ/kg-K). Find and write the equation for c_v in kJ/kg-K. Then, using the equations, calculate the values of c_p and c_v at $T = 1100$ K. Also calculate γ at 1100 K.